CENG381 Stochastic Processes -- Mixed Exam

Mixed Exam 2 -- Answer Key

Q1 Solution (30 points)

a) Row sums and holding times:

Row 0: -2.0 + 1.5 + 0.5 = 0 (OK)

Row 1: 1.0 - 3.0 + 2.0 = 0 (OK)

Row 2: 0.5 + 1.5 - 2.0 = 0 (OK)

```
q(0) = 2.0   =>   H(0) ~ Exponential(2.0),   E[H(0)] = 0.5 time units
q(1) = 3.0   =>   H(1) ~ Exponential(3.0),   E[H(1)] = 1/3 time units
q(2) = 2.0   =>   H(2) ~ Exponential(2.0),   E[H(2)] = 0.5 time units
```

State 1 (Busy) has the shortest expected holding time.

b) Stationary distribution from pi * Q = 0:

Balance equations (rate in = rate out for each state):

```
2.0*pi(0)  =  1.0*pi(1) + 0.5*pi(2)   ... (I)
3.0*pi(1)  =  1.5*pi(0) + 1.5*pi(2)   ... (II)
2.0*pi(2)  =  0.5*pi(0) + 2.0*pi(1)   ... (III)
```

From (II): pi(1) = 0.5*pi(0) + 0.5*pi(2) ... (II')

Sub into (I): 2*pi(0) = 0.5*pi(0) + 0.5*pi(2) + 0.5*pi(2)

2*pi(0) = 0.5*pi(0) + pi(2) => pi(2) = 1.5*pi(0)

From (II'): pi(1) = 0.5*pi(0) + 0.5*(1.5*pi(0)) = 0.5*pi(0) + 0.75*pi(0) = 1.25*pi(0)

Normalise: pi(0) + 1.25*pi(0) + 1.5*pi(0) = 1 => 3.75*pi(0) = 1

```
pi(0) = 4/15,   pi(1) = 5/15 = 1/3,   pi(2) = 6/15 = 2/5
```

Check: 4/15 + 5/15 + 6/15 = 15/15 = 1 (OK)

c) Detailed balance test between states 0 and 2:

```
pi(0) * Q(0,2) = (4/15) * 0.5  = 2/15
pi(2) * Q(2,0) = (6/15) * 0.5  = 3/15
```

2/15 != 3/15, so detailed balance does NOT hold between 0 and 2.

The chain is NOT reversible.

Q2 Solution (35 points)

a) rho, pi(n), P(N > 3):

```
rho = lambda/mu = 5/8 = 0.625  <  1  =>  queue is stable
pi(n) = (1 - rho) * rho^n = 0.375 * (0.625)^n
pi(0) = 0.375  (server idle 37.5% of the time)
```

P(more than 3 customers) = P(N >= 4) = rho^4:

```
P(N >= 4) = rho^4 = (0.625)^4 = 0.1526  (about 15.3%)
```

b) E[N], E[T], E[T_q]:

```
E[N] = rho / (1 - rho) = 0.625 / 0.375 = 5/3 ~ 1.667 customers
E[T] = E[N] / lambda = (5/3) / 5 = 1/3 hour = 20 minutes
E[T_q] = E[T] - 1/mu = 1/3 - 1/8 = 8/24 - 3/24 = 5/24 hour ~ 12.5 minutes
```

c) Server vacation effect (qualitative):

Waiting to accumulate 2 customers before starting service means customers who arrive when 0 or 1 are in the system must wait even though the server is idle. This increases the average waiting time and E[N] compared to the standard M/M/1 where service begins immediately.

Q3 Solution (35 points)

a) P^2(0,0) via Chapman-Kolmogorov:

```
P^2(0,0) = P(0,0)*P(0,0) + P(0,1)*P(1,0)
         = 0.6*0.6 + 0.4*0.3
         = 0.36 + 0.12 = 0.48
```

b) Eigenvalues of P:

With p = P(0,1) = 0.4 and q = P(1,0) = 0.3:

```
lambda(1) = 1
lambda(2) = 1 - p - q = 1 - 0.4 - 0.3 = 0.3
```

c) P^n(0,0) using diagonalization:

```
P^n(0,0) = q/(p+q) + lambda(2)^n * p/(p+q)
         = 0.3/0.7 + (0.3)^n * 0.4/0.7
         = 3/7 + (4/7) * (0.3)^n
```

For n = 3:

```
P^3(0,0) = 3/7 + (4/7)*(0.3)^3 = 3/7 + (4/7)*0.027
         = 3/7 + 0.108/7 = (3 + 0.108)/7 = 3.108/7 ~ 0.444
```

As n -> infinity: (0.3)^n -> 0, so:

```
P^n(0,0) -> 3/7 ~ 0.429 = pi(0) (stationary probability of state 0)
```